is just a minute, infinitesimal mote of dust; on the other hand how peculiar it must seem that we, tiny creatures on a tiny planet, are nevertheless capable of knowing though not the essence at least the existence and the dimensions of the basic building blocks of the entire great Cosmos!

But this is still not the end of the wonder of it. Physical research has established as an incontestable fact that these basic building blocks of the Universe do not exist unrelated in isolated groups, but that all of them are mutually interlinked according to one uniform plan. In other words, every process in nature is subject to a universal and up to a point knowable law.

I want to mention at this place just one example: The law of the conservation of energy. There are various forms of energy in nature—kinetic energy, the energy of gravity, heat, electricity, magnetism. All the energies together form the energy supply of the world. The quantity of this energy supply is constant; it cannot be increased or diminished by any process in nature. All changes in nature are in reality simply the transformations of one form of energy into another. For instance, when kinetic energy is lost by friction, an equivalent quantity of thermal energy results.

The law of the conservation of energy is valid in every branch and field of physics, both according to the classical theory and to quantum mechanics. To be sure, there have often been attempts to challenge its precise validity in connection with the processes taking place within a single atom, and to assign to it a mere statistical significance for such processes. But every single experiment thus far undertaken has shown that such attempts are futile, and there is no reason to deny that the law of the conservation of energy is an absolutely and universally valid law of nature.

Those with positivistic leanings counter frequently with the critical objection that there is nothing extraordinary about the universal validity of such a law. According to them the explanation of the mystery is simply that after all it is man himself who prescribes the laws for nature. And in claiming this, they even cite Immanuel Kant in support of this view.

However, as I have pointed it out, the laws of nature were not invented by man, but external factors forced him to recognize them. An *a priori* approach to the laws of nature, as well as to the universal constants, could make us imagine them quite differently from what they are in reality. But the positivist reference to Kant is based on a gross misunderstanding. Kant did not teach that man actually prescribes laws for nature. He taught simply that whenever man formulates the laws of nature, he always adds something of his own, too. Otherwise how would it be conceivable that according to Kant’s own statement, no external impression inspired in him a more profound feeling of respectful humility than the sight of the starry skies? After all, respectful humility is not exactly the attitude which a man is in the habit of assuming toward a rule formulated by himself. Obviously, such sentiment is alien to the